Smoothing Splines Quiz

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Instructions

1. What is a smoothing spline? **1 / 1 point**

- ○ A spline that passes through every data point.
- ○ A spline exclusively used for interpolating missing data.
- ○ A spline that removes all noise from a dataset.
- ◉ A spline used to fit a set of data points in a way that ensures smoothness while controlling overfitting.

Correct
Correct! Smoothing splines balance fitting the data closely while maintaining smoothness.

2. How do smoothing splines differ from interpolating splines? **1 / 1 point**

- ○ Smoothing splines cannot be used for data with more than two dimensions.
- ◉ Smoothing splines do not necessarily pass through all the data points, unlike interpolating splines.
- ○ Smoothing splines use only linear functions.
- ○ Smoothing splines are not continuous.

Correct
Correct! Smoothing splines aim for a smooth curve, not necessarily passing through every point.

3. What role does the smoothing parameter play in a smoothing spline? **1 / 1 point**

- ◉ It controls the trade-off between the spline's smoothness and its closeness to the data points.
- ○ It represents the number of knots in the spline.
- ○ It determines the degree of the polynomial used in the spline.
- ○ It adjusts the color and style of the spline when plotted.

Correct
Correct! The smoothing parameter balances the fit to the data and the smoothness of the spline.